

Anika Sharma

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Research interests

Interpretability of LLMs, HCI, Machine Learning, Social Natural Language Processing, AI Alignment and Safety

Education

2022 – 2026

University of North Carolina at Chapel Hill – Chapel Hill, North Carolina

BS in Computer Science and BS in Data Science (*Graduated with distinction and highest honors, GPT: 3.61*)

Honors Thesis Title: Computational Approaches to Measuring Latent Human Constructs

Advisor: Professor Neil Gaikwad, Professor Shashank Srivastava

Selected coursework

- *Machine Learning and CS*: Systems programming, Organization of Code, Computer Organization, Models of Languages and Computation, Algorithms and Analysis, Machine Learning, Video Recognition, Foundations of SWE, NLP, Deep Learning, Deep Generative Models, Robotics, Human Computer Interaction
- *Math and Statistics*: Probability for Data Science (Stor 535), Applied Linear Algebra (Math 347), Quantitative Finance (Math 590)

Honors and Awards

2026

Highest Honors, Department of Computer Science

Defended my thesis on computational approaches to measuring latent human constructs with highest honors..

2023, 2025

Grace Hopper Scholar, UNC Department of CS

Competitive institutional scholarship awarded by the CS department to support selected students' attendance at the Grace Hopper Celebration, based on academic excellence, leadership in computing, and contributions to diversity and inclusion in tech.

2025

SARC Summer Research Scholar, UNC Department of CS

Awarded to 10–15 students university-wide to support faculty-mentored summer research. Provided full tuition funding for intensive research coursework and project development.

2023-2025

Dean's List, UNC–Chapel Hill

Recognized for high academic performance (3.5+ GPA) across multiple semesters.

2025

Data Science Travel Grant, UNC School of Data Science and Society

Funded to present first-author research on LLM representations of abortion stigma at the Georgia Tech Responsible AI Summit. Award supported travel and registration.

Publications

- 2025 **Can LLMs Understand What We Cannot Say? Measuring Multilevel Alignment Through Abortion Stigma Across Cognitive, Interpersonal, and Structural Levels**
Anika Sharma, Malavika Mampally, Chidaksh Ravuru, Kandyce Brennan, Neil Gaikwad.
Pre-print available (arXiv), Under review at CHI 2027.
- 2025 **Adversarially Probing Cross-Family Sound Symbolism in 27 Languages**
Anika Sharma, Tianyi Niu, Emma Wrenn, Shashank Srivastava.
Accepted into EMNLP main 2026, Oral Presentation, Pre-print available (arXiv).
- 2026 **"Girl, I'm so Serious": CARE, a Capability Framework for Reproductive Equity in Human-AI Interaction**
Alice Zhong, Phoebe Chen, Anika Sharma, Kandyce Brennan, Neil Gaikwad.
Accepted as an extended abstract/poster at CHI'26; Equal contributor.
- 2026 **CARE: A Capability-Based Measurement Framework for Reproductive Equity in Human-AI Interaction**
Alice Zhong, Phoebe Chen, Punya Aragula, Anika Sharma, Kandyce Brennan, Neil Gaikwad.
- 2026 **Computational Approaches to Measuring Latent Human Constructs**
Anika Sharma.
Honors thesis defended with highest distinction.

Research experience

March 2023 –
Present

Language and Learning Lab

Advisor: Professor Shashank Srivastava

I am researching interpretability and social reasoning in language models through cross-linguistic analysis and multi-agent simulations.

- **Cross-Linguistic Sound Symbolism in Size Adjectives** — Built a 27-language phonological dataset (810 adjectives) and developed a cross-lingual ML framework combining Transformer pretraining, leave-one-language-out transfer, and gradient-reversal adversarial learning to isolate sound–meaning representations from genealogical and language-specific signals. First-author paper accepted to EMNLP 2026 Main Conference (Oral Presentation).
- **Gender Bias in LLM Social Judgment** — Designed a gender-counterfactual evaluation of social judgment across four LLMs and 19 cultural personas, using 766 naturalistic interpersonal conflicts to disentangle leniency toward female protagonists from protectiveness toward female counterparts. Found that favorable gender bias is driven more consistently by harsher judgments when the affected party is a woman than by leniency toward women being judged. First-author paper under review at ARR.

July 2025 – Present

Society-Centered AI Lab

Advisor: Professor Neil Gaikwad

I research human-AI sensemaking, fairness, and how hidden assumptions in AI systems shape user reasoning and decision-making.

- **Behavioral Coherence in Sensitive-Domain LLMs** — Developed a human-centered evaluation framework for surfacing latent social assumptions across LLM interactions, combining validated psychometric measures, controlled demographic personas, and expert review. Across five LLMs, identified systematic demographic inference and recurring interactional defaults around judgment, secrecy, and support, motivating design guidance for context-sensitive reproductive-health AI. First-author paper under review at CHI 2027.
- **CARE Framework for Reproductive Health AI** — Developed a capability-based normative framework, grounded in Amartya Sen’s capability approach, for evaluating how sexual and reproductive health technologies expand or constrain users’ reproductive freedoms. Applied the framework to conversational AI systems to identify epistemic and interactional harms, including limited source transparency, rigid responses, and constraints on user agency. Published as an extended abstract at CHI 2026; long-form paper available on arXiv.
- **Interactive AI Fairness & Model-Building Education** — Developed an interactive platform for teaching fairness-aware machine learning in high-stakes settings, using AST-based program analysis and a Scratch-style visual interface to surface hidden defaults and assumptions in AI-generated code. Designed Socratic tutoring and guided model-building workflows that help students interrogate modeling choices, compare downstream fairness outcomes, and reason about tradeoffs rather than passively accepting AI-generated decisions. Manuscript in preparation.

Teaching experience

Spring 2026

Teaching assistant, Comp 488: Data Science for Business (UNC)

Comp 488 is a higher division elective for the CS and Data Science major. We go over the fundamentals of cloud computing and AI through exploring real world, unstructured datasets.

Fall 2025

Teaching assistant, Data 520: Research-Methods for Socially Responsible AI: An Ethical Expedition (UNC)

Data 520 is a required course for the AI/ML concentration in Data Science and CS. Mentored 40+ undergraduate students developing AI systems for ecological decision-making in partnership with the U.S. Embassy Kathmandu, South Asia Regional Environment Office (REO). Guided teams in designing culturally aligned, region-specific LLM applications that generate landscaping recommendations for the Indo-Gangetic plains.

Spring
2025–Present

Coach & Judge, VEX Robotics
Coach for a high school VEX Robotics team, teaching students to program autonomous and driver-controlled robots in C++ with sensor integration and PID control. Served as an official VEX Robotics Competition judge evaluating engineering notebooks and team interviews.

Industry experience

Summer 2024

Gain Life (AI/Machine Learning Intern) – Remote/Boston, MA
Automated medical-claim risk assessment using multi-threaded Bedrock API pipelines, reducing categorization time by 93% and improving risk prediction accuracy by 7.5% with custom XGBoost modeling. Built OCR workflows using DocumentAI, DocTR, and Tesseract for form processing, and applied nearest-neighbors analysis to identify high-risk demographic segments.

Service

2023 – 2025

Software Engineering Lead, SAILea
Developed a full-stack web platform for SAILea, a nonprofit supporting AI/ML education for high school clubs nationwide. Built features for hosting educational materials, interactive lessons, and follow-along activities, with authentication, role-based access control, user dashboards, and a forthcoming real-time chat system. Built with the MERN stack (MongoDB, Express.js, React, Node.js). Now serving in an advisory capacity to SAILea's high school leadership board, mentoring students as they take over development and program decisions.

2024 – 2026

Lead Student Experience Ambassador, UNC Department of Computer Science
Managed a team of 10+ Student Experience Ambassadors and lead programming that supports historically excluded identities in CS, including heritage celebrations and corporate partner outreach. Served as the sole undergraduate member of the department's Student Experience Committee, co-leading resource development efforts and contributing to policy recommendations from the annual Climate Survey.

Technical skills

Programming languages

Proficient in: Python, Java, JS, TS

Familiar with: C, C++, R

Machine Learning / AI

PyTorch, TensorFlow, scikit-learn, XGBoost, Computer Vision, Deep Learning

Other interests

I love to run (running the charlotte half marathon this November), swim and read!